

Quantitative Analysis of Media Bias and Stock Price Dynamics: The 2020 Shock

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Abstract—This paper studies how media reporting about large-cap firms changed around the 2020 recession and whether shifts in stance of news outlets towards these firms predict firm-level stock outcomes. Using a NewsMTSC pipeline to compute daily target-dependent stance from 690,586 relevance-filtered full-text article headlines (2015–2025) across 26 S&P 500 firms, we align stance with firm-day returns and market controls and address three research questions: whether media stance towards firms significantly shifted after January 2020 (RQ1), whether firm-idiosyncratic returns shifted (RQ2), and whether stance predicts returns in a dynamic sense (RQ3). RQ1 and RQ2 use two-way fixed-effects pre/post panel regressions; RQ3 uses a weekly vector autoregression with Granger-causality tests. We find no significant aggregate level shift in either stance or returns once common daily and monthly shocks are absorbed, or a dynamic stance–return link that is firm-specific rather than a market-wide regularity.

I. INTRODUCTION

The 2020 recession was a natural breakpoint for studying the interaction between media narratives and financial markets. COVID-era financial news coverage expanded in volume and intensity, and a growing body of evidence documents a structural change in news-market dynamics during and after the shock. This paper addresses the broad research question: *Did the tone of firm-specific media reporting shift after 2020, and if it did, did media tone have a significant impact on the market?*

We approach the problem with a target-dependent stance framework rather than generic sentiment. NewsMTSC [cite], a library for aspect based sentiment analysis, allows us to score whether an article title is positive, negative, or neutral toward a specific firm even when multiple firms are mentioned in the same text. While a general sentiment score cannot distinguish positive coverage of one firm from negative coverage of another in the same article. Target-dependent stance makes it feasible to construct a daily, firm-level media stance signal and evaluate how it behaves across time and shocks.

The study is grounded in three observations. The 2020 shock is widely documented as a period in which media coverage and market behavior changed in a persistent way [cite]. This study attempts to discover a correlation, if any, between the two.

Second, the data pipeline needs to be precise: naive keyword search introduces many false positives and inflates noise in the stance signal. Third, if stance is to be linked to returns in a credible way, the design must rule out coverage volume effects, pre-trend differences, and spurious correlations driven by macro shocks.

We built a three-stage pipeline for this analysis: (i) relevance classification that filters articles to consider only financial news related articles that actually report on the target firm, (ii) NewsMTSC stance scoring with confidence thresholds to extract the stance of the titles towards the target firm, (iii) two-way fixed-effects pre/post panel regressions (RQ1, RQ2) with a weekly vector autoregression and Granger-causality tests (RQ3) to characterize both the level shifts and the dynamic stance–return channel.

The remainder of the paper proceeds as follows. Section II states the research questions and formal hypotheses. Section III describes the data pipeline. Section IV develops the methodology and estimating equations. Section V presents the regression results for RQ1 and RQ2. Section VI presents the VAR and Granger-causality analysis for RQ3. Section VII addresses identification and robustness, and Section VIII concludes.

II. RESEARCH QUESTIONS AND HYPOTHESES

The three research questions addressed through this study are as follows:

RQ1: Stance shift. Did the tone of firm-specific media coverage shift significantly after January 2020?

$$H_0^{(1)} : \beta = 0$$

$$H_1^{(1)} : \beta \neq 0$$

where β is the coefficient on Post_t in the two-way fixed-effects stance regression (Section IV-C).

RQ2: Returns shift. Did firm-idiosyncratic daily returns shift after January 2020 beyond market co-movement?

$$H_0^{(2)} : \beta = 0$$

$$H_1^{(2)} : \beta \neq 0$$

where β is the coefficient on Post_t in the two-way fixed-effects returns regression.

RQ3: Stance-return predictability. Does past media stance Granger-cause future returns, and did that relationship change post-2020?

$$H_0^{(3)} : a_1^{(2,1)} = \dots = a_p^{(2,1)} = 0$$

$$H_1^{(3)} : \exists \ell \text{ s.t. } a_\ell^{(2,1)} \neq 0$$

where $a_\ell^{(2,1)}$ are the lagged-stance coefficients in the return equation of the VAR.

III. DATA CONSTRUCTION

A. Corpus and Coverage

We begin with a large corpus of scraped full-text news articles (2015–2025) covering large-cap firms that are consistently observable across the window, drawn from the S&P 500. Each record stores the publication, timestamp, URL, title, first paragraph, and full body text. The raw corpus contains 4,706,589 firm-article links (Table I); coverage is highly skewed, with a handful of firms dominating and others sparsely covered.

B. Data Preprocessing

The raw data obtained through the scraping pipeline is further preprocessed in three steps: title-level de-duplication, relevance filtering (Section IV-A), and target-dependent stance scoring (Section IV-B). All downstream analyses were run on the resulting scored table, which contains 690,586 high-confidence firm-article stance observations across 26 firms. Collapsing to the firm-day level yields the 63,200-observation stance panel used for RQ1; aligning stance to firm-day market data yields the 62,667-observation return panel used for RQ2. Table II summarizes the funnel.

Figure 1 shows article coverage per firm in the scored analysis sample. The distribution spans more than four orders of magnitude—from AT&T (139,701 scored articles) down to Eaton (5)—confirming that coverage is concentrated in a subset of heavily reported firms, which the firm fixed effects in Section IV-C absorb.

IV. METHODOLOGY

A. Relevance Filtering and Annotation

A company name in a headline does not guarantee that the article reports on that company as a corporate entity. “Apple” may refer to a product or appear only in passing, and generic personal stories and experiences around a product routinely name a firm without covering financial details about it. To remove such noise, we keep only those articles in which the firm is the primary subject of corporate/financial events: earnings, mergers and acquisitions, executive changes, large layoffs, regulatory or legal action, major product launches, or credit and bankruptcy events. This required us to develop a relevance filter that selects only financial articles around firms for further analysis.

The filter has two stages, a title-level pre-filter that requires the company name to appear in the headline, followed by a

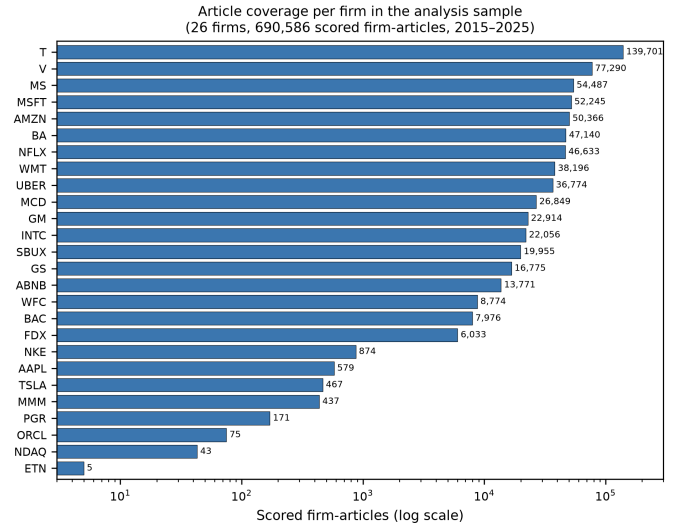


Fig. 1. Scored firm-articles per firm in the analysis sample (26 firms, 690,586 scored firm-articles from `articles_no_title_deduped`, 2015–2025; log scale).

supervised classifier that judges materiality. Figure 2 summarizes the full stage.

We manually labeled 750 article titles under this rubric, of which 227 are relevant and 523 are irrelevant, forming the training data for the classifier. A separate held-out test set of 190 titles is used as the test data. The classifier is a DeBERTa-v3-base model fine-tuned on the training titles. Trained on the labeled set alone it reached a precision of 0.65 at a recall of 0.71 (Table III). The binding constraint is the small label budget together with the skew between the labeled relevant rate, near 30%, and the true rate in the corpus, which is closer to 10 to 15%.

To sharpen the decision boundary without further manual labeling, we add 10,025 synthetic titles. These cover the patterns the base model most often confuses, including consumer promotions, industry overviews, immaterial lawsuits, brand comparisons, and articles that name a firm only in passing, together with harder examples of genuine material events phrased in unusual ways. Training on the combined set with focal loss and a fivefold weighting of the labeled examples raises precision from 0.65 to 0.88 while holding recall above 0.85.

Table III compares the configurations on the held-out test set. A three-model ensemble with rule-based post-processing reaches the highest precision at 0.94, but we adopt a single consolidated DeBERTa-v3 model: it matches the ensemble on F1, 0.87 against 0.87, using one classifier rather than an ensemble and a separate rule layer, which is simpler to reproduce and to report. At its operating threshold the model attains a precision of 0.88, a recall of 0.86, and an F1 of 0.87; a stricter threshold raises precision to 0.92 at a recall of 0.81 (Figure 3). Applied to the full corpus of 695,731 scored articles, it labels 90,579 as relevant, or 13%, consistent with the expected base rate. These relevant articles are then

TABLE I
YEARLY COUNT OF ARTICLES IN THE RAW CORPUS (2015–2025); THESE SUM TO THE 4,706,589 STARTING LINKS.

Year	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Total articles	570,999	625,582	652,741	525,259	559,027	363,962	291,144	260,115	204,281	204,007	449,472

TABLE II
SAMPLE CONSTRUCTION FUNNEL FROM RAW CORPUS TO ESTIMATION PANELS

Stage	Count
Raw corpus (firm-article links, 2015–2025)	4,706,589
Scored firm-articles (relevance-filtered, NewsMTSC)	690,586
Firms with scored coverage	26
RQ1 stance panel (firm–days)	63,200
RQ2 return panel (firm–days)	62,667

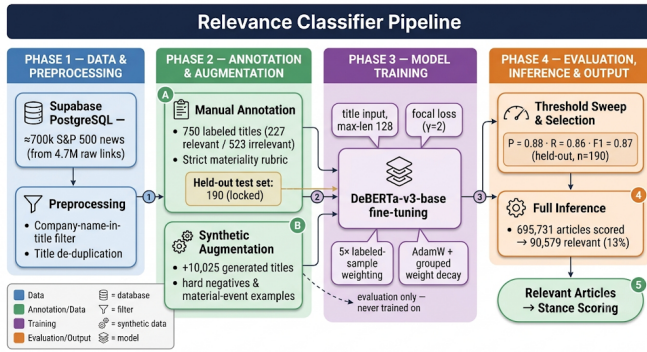


Fig. 2. Relevance classifier pipeline: raw S&P 500 news titles, preprocessing, manual annotation and synthetic augmentation, DeBERTa-v3 fine-tuning, threshold selection on a held-out test set, and full-corpus inference.

TABLE III
RELEVANCE CLASSIFIER PERFORMANCE ON THE HELD-OUT TEST SET ($n = 190$), EACH MODEL AT ITS F1-OPTIMAL THRESHOLD. THE ENSEMBLE IS SHOWN FOR REFERENCE; THE CONSOLIDATED SINGLE MODEL IS ADOPTED.

Training strategy	P	R	F1
Labeled titles only	0.65	0.71	0.68
+ synthetic (broad)	0.78	0.76	0.77
+ synthetic (targeted, focal)	0.86	0.88	0.87
Consolidated single model	0.88	0.86	0.87
Ensemble (reference)	0.94	0.81	0.87

fed to the stance classification stage as input as described in Section IV-B.

B. Stance Scoring

Stance is computed with the NewsMTSC model for target-dependent stance. For each qualifying article it returns probabilities over positive, negative, and neutral stance toward the aspect, which is the name of the firm in this case. The baseline specification keeps all scored articles (confidence threshold $\tau = 0$) to maximize coverage; we report sensitivity to higher-

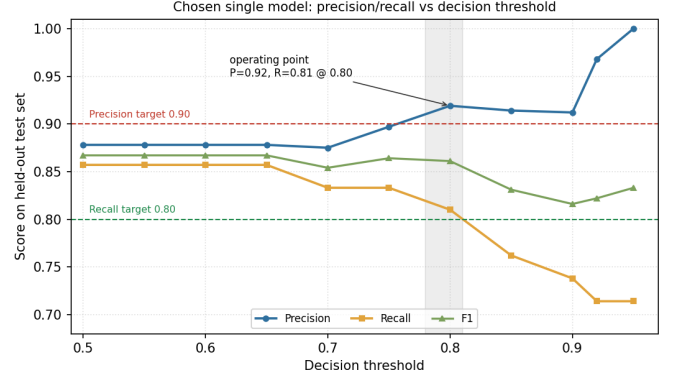


Fig. 3. Precision, recall, and F1 of the chosen single model on the held-out test set as a function of the decision threshold. A threshold near 0.80 satisfies both operating targets, reaching precision 0.92 at recall 0.81, which is why this single classifier is adopted in place of the ensemble.

confidence subsets ($\tau = 0.6, 0.9$). Daily firm-level stance is the article mean,

$$\text{stance}_{i,t} = \frac{1}{N_{i,t}} \sum_{a \in A_{i,t}} (p_a^+ - p_a^-), \quad (1)$$

where $A_{i,t}$ is the set of scored articles for firm i on day t and $N_{i,t}$ is the count. We also retain article volume as a covariate to separate tone shifts from coverage shifts.

C. Market Alignment and Controls

The stance series is aligned to firm-day market outcomes: daily returns, the S&P 500 index return, and VIX, drawn from `stock_prices` and the VIX daily series. This alignment yields the panels used in the RQ1–RQ3 specifications.

D. Identification and Panel Design

Both RQ1 and RQ2 use a pre/post design on a firm–day panel. The treatment indicator

$$\text{Post}_t = \mathbf{1}[t \geq 2020-01-01] \quad (2)$$

partitions the sample into a pre-period (2015–2019) and post-period (2020–2025). Firm fixed effects α_i absorb permanent cross-firm heterogeneity; time fixed effects δ_t absorb common shocks at the day or month level depending on the outcome. The stance outcome is the signed daily mean,

$$\text{bias}_{it} = \frac{1}{|A_{it}|} \sum_{a \in A_{it}} (p_a^+ - p_a^-), \quad (3)$$

and the returns outcome is the log return $\text{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$. Both equations are estimated by within-transformation OLS with HC1-robust standard errors; day-clustered standard errors are additionally reported for RQ1.

E. RQ1: Stance Regression

$$\text{bias}_{it} = \alpha_i + \delta_t^{\text{day}} + \beta \text{Post}_t + \gamma \text{vol}_{it} + \varepsilon_{it} \quad (4)$$

Firm effects α_i absorb each firm's structural coverage level; daily time effects δ_t^{day} absorb market-wide sentiment common to all firms on a given day. The volume control $\text{vol}_{it} = |\mathcal{A}_{it}|$ is required because the mean of a noisy signal mechanically concentrates as more articles are averaged, so volume must be held fixed to isolate a genuine tone shift from a coverage shift. Under $H_0^{(1)}$, $\beta = 0$.

F. RQ2: Returns Regression

$$\text{return}_{it} = \alpha_i + \delta_t^{\text{month}} + \beta \text{Post}_t + \gamma_1 \text{sp500_ret}_t + \gamma_2 \text{vix}_t + \varepsilon_{it} \quad (5)$$

Monthly time effects δ_t^{month} absorb common market-level shocks at the monthly frequency, which is the resolution at which macro regime changes operate. The S&P 500 return and VIX are included as daily controls within each month. Under $H_0^{(2)}$, $\beta = 0$.

G. RQ3: VAR and Granger Causality

For the dynamic analysis we estimate a reduced-form bivariate VAR of order p on the endogenous vector $\mathbf{y}_t = (\Delta \text{bias}_t, r_t)^\top$:

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^p \mathbf{A}_\ell \mathbf{y}_{t-\ell} + \mathbf{u}_t \quad (6)$$

where Δbias_t is the first-differenced weekly bias index (required for stationarity; see Section VI) and r_t is the weekly log return. The $H_0^{(3)}$ restriction is $a_1^{(2,1)} = \dots = a_p^{(2,1)} = 0$, tested via a joint Wald F -test with Newey–West HAC standard errors.

V. RESULTS

A. RQ1: No Significant Aggregate Stance Shift

Table IV(a) reports estimates from Eq. (4) on $n = 63,200$ firm–day observations (26 firms, 4,018 days). We fail to reject $H_0^{(1)}$: the post-shock coefficient is $\hat{\beta} = +0.0141$ with HC1 SE = 0.0136 ($p = 0.30$); day-clustered inference is nearly identical (SE = 0.0134, $p = 0.29$), confirming that within-day cross-firm dependence is not driving the result. Article volume is strongly negative (-0.00097^{***}): heavier-coverage firm-days regress toward neutral tone, validating the volume control. The within $R^2 = 0.0017$ reflects the well-known dominance of idiosyncratic article noise at the firm-day level.

The null result is informative. A joint F -test of the 4,017 daily effects is strongly significant ($F = 1.135$, $p = 1.1 \times 10^{-8}$), meaning common daily shocks drive most of the time variation in stance. Once those shocks are absorbed by δ_t^{day} , the static Post step carries almost no residual identifying variation—the day fixed effects effectively span it. We therefore fail to detect a systematic within-firm stance shift beyond what the broader news environment already explains.

B. RQ2: No Significant Firm-Idiosyncratic Return Shift

Table IV(b) reports estimates from Eq. (5) on $n = 62,667$ firm–day observations (26 firms; pre-period mean returns tightly clustered, range 0.0019, SD 0.0004). We fail to reject $H_0^{(2)}$: once monthly time effects absorb common market-regime variation, the post-shock coefficient is $\hat{\beta} = -0.0018$ (SE = 0.0142, $p = 0.90$). The S&P 500 return regressor is absorbed by the month fixed effects ($\hat{\gamma}_1 \approx 0$, $p = 0.90$), and VIX is similarly uninformative within month ($p = 0.93$). The within $R^2 \approx 0$ follows directly: with the dominant systematic factor (index returns) loaded entirely onto the monthly FE, the regressors have negligible residual explanatory power.

Together, RQ1 and RQ2 establish a clean null baseline: after properly absorbing common daily and monthly shocks, neither media stance nor firm-level returns show a detectable aggregate level shift around January 2020. This motivates the dynamic question in Section VI—whether the *lead–lag relationship* between stance and returns changed post-shock, even if the marginal distributions did not.

VI. VECTOR AUTOREGRESSION AND GRANGER CAUSALITY ANALYSIS

The regressions in the previous section ask whether the level of each series shifts after 2020. They are silent on the dynamic relationship between media stance and returns: whether past stance helps predict future returns, whether returns feed back into coverage, and whether that lead–lag structure itself changed after the shock. We answer these questions with a vector autoregression (VAR) and Granger-causality framework, supported by an eight-stage data pipeline that constructs a clean, stationarity-validated weekly panel before any dynamic model is fit. We move to a weekly frequency for this analysis because daily firm-level stance is sparse and noisy for several firms, whereas weekly aggregation yields adequate coverage (median 92.9% of weeks across firms) while retaining enough observations (≈ 522 aligned weeks per firm) for lag-rich models. Per-firm results for every pipeline stage are tabulated in the Appendix.

A. The VAR Model

For a firm with endogenous vector $\mathbf{y}_t = (\Delta \text{bias}_t, r_t)^\top$, where Δbias_t is the first difference of the weekly bias index and r_t is the weekly log return, a reduced-form VAR of order p is

$$\mathbf{y}_t = \mathbf{c} + \sum_{\ell=1}^p \mathbf{A}_\ell \mathbf{y}_{t-\ell} + \mathbf{u}_t, \quad (7)$$

with coefficient matrices $\mathbf{A}_\ell \in \mathbb{R}^{2 \times 2}$ and innovations $\mathbf{u}_t$ having covariance Σ . The off-diagonal entries of the $\mathbf{A}_\ell$ carry the cross-dynamics: a nonzero (1, 2) entry means past returns move current stance, and a nonzero (2, 1) entry means past stance moves current returns. The VAR is the minimal model that treats both variables as endogenous and lets the data, rather than an a priori causal ordering, reveal the direction of predictability, which is precisely what a media–market feedback question requires.

TABLE IV

REGRESSION RESULTS FOR RQ1 (STANCE SHIFT, EQ. 4) AND RQ2 (RETURNS SHIFT, EQ. 5). BOTH SPECIFICATIONS FAIL TO REJECT THE NULL AT CONVENTIONAL LEVELS ONCE COMMON DAILY (RQ1) AND MONTHLY (RQ2) SHOCKS ARE ABSORBED.

(a) RQ1 — Daily Stance			
Firm + day FE, HC1 SE			
Variable	Coef.	SE	<i>p</i>
Post ($\hat{\beta}$)	+0.01413	0.01362	0.300
Article volume	−0.00097	0.00008	< 0.001
<i>n</i> = 63,200; 26 firms; 4,018 days			
Within R^2 = 0.0017			

(b) RQ2 — Daily Returns			
Firm + month FE, HC1 SE			
Variable	Coef.	SE	<i>p</i>
Post ($\hat{\beta}$)	−0.00184	0.01424	0.897
S&P 500 return	≈ 0	—	0.902
VIX	+0.00061	0.00654	0.925
<i>n</i> = 62,667; 26 firms			
Within R^2 ≈ 0			

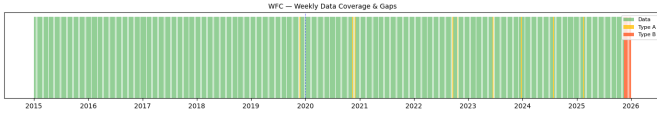


Fig. 4. Weekly data-coverage and gap map for Wells Fargo (WFC), the lowest-coverage retained firm. Short and medium gaps are imputed; the series still clears the $\geq 60\%$ coverage threshold.

B. Stage 1–4: Building the Weekly Panel

The pipeline begins with a *data inventory* (Stage 1) that audits, for every candidate firm, article counts, coverage, and the share of zero-article days, and flags sparse firms (after relevance filtering the median firm posts an article on fewer than half of all trading days) for exclusion. This audit motivates the move to a weekly frequency and the retention of the sixteen firms that clear the coverage threshold. Stage 2 constructs the *weekly bias index* by aggregating signed article stances into ISO weeks and verifies distributional quality (mean, skew, kurtosis, and the incidence of single-article weeks). Stage 3 performs *gap analysis and imputation*: weekly gaps are classified by length, with short Type-A gaps (1–2 weeks) forward-filled and medium Type-B gaps (3–8 weeks) linearly interpolated, while long Type-C gaps are left missing; 1,131 weekly observations are imputed (822 forward-fill, 309 interpolation) and all sixteen firms clear a $\geq 60\%$ coverage threshold with no gap exceeding eight weeks (the sparsest, Airbnb and Wells Fargo, near 69–71%). Crucially, both an imputed and an observed-only series are retained so that every downstream test can be re-run to confirm results are not artifacts of imputation. Figure 4 illustrates the coverage map for the sparsest retained firm.

Stage 4 constructs the *returns series* as weekly log returns from Monday-anchored ISO weeks and aligns them to the bias index. Log returns are used because they are additive across time and approximately symmetric, which suits a linear dynamic model. Quality checks flag extreme weekly moves ($> \pm 15\%$, concentrated in the March 2020 crash window) for later winsorization, and every firm clears the minimum of 100 aligned weeks, yielding a merged panel of 8,090 firm-week rows.

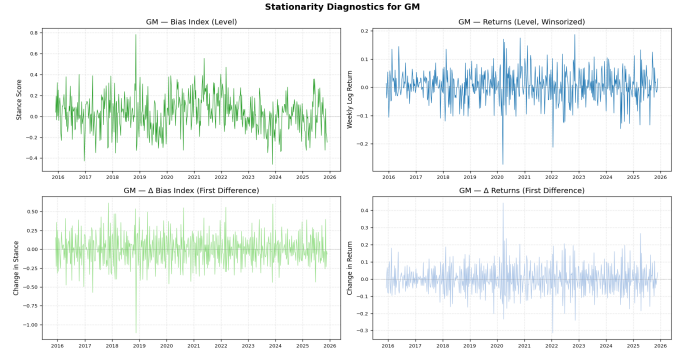


Fig. 5. Stationarity diagnostics for General Motors (GM): the bias index and weekly return in levels (left) and first differences (right), with ADF and KPSS verdicts driving the integration-order classification.

C. Stage 5: Stationarity and Cointegration

A VAR in levels is only valid if the series are stationary or jointly cointegrated; otherwise the regression is spurious. Each series is classified by combining the Augmented Dickey–Fuller (ADF) test, whose null is a unit root, with the KPSS test, whose null is stationarity. A series is called $I(0)$ when ADF rejects ($p < 0.05$) and KPSS does not, $I(1)$ in the opposite case, and “conflicting” (a likely structural break) when both reject. As expected for financial data, weekly log returns are $I(0)$ for essentially every firm, while on the relevance-filtered series the bias index is $I(0)$ in levels for about half the firms and $I(1)$ or break-driven for the rest. To treat all firms under one specification and avoid a spurious levels regression where bias is $I(1)$, the VAR is specified on $(\Delta \text{bias}_t, r_t)$ rather than on the levels: differencing the bias index renders it stationary everywhere. A Johansen trace test detects a cointegrating relation for two firms, Boeing (trace = 367.2) and Wells Fargo (trace = 317.5), both far above the 95% critical value of 15.5, which would otherwise call for a VECM; these are flagged as robustness exceptions to the differenced specification. Figure 5 shows the level-versus-difference diagnostic for a representative firm.

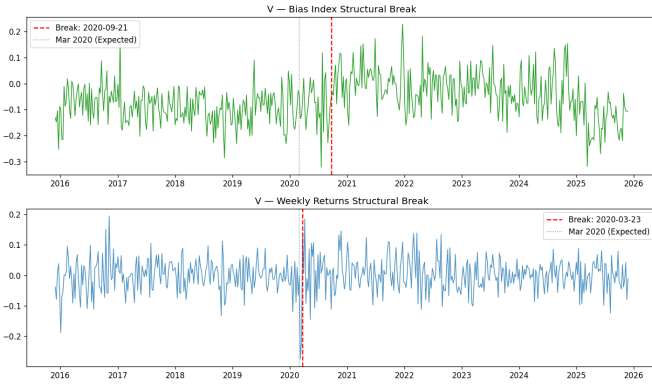


Fig. 6. Bai-Perron structural breaks for Visa (V). The return break (2020-03-23) coincides with the COVID crash; the bias-mean break (2024-06-10) is firm-specific and later.

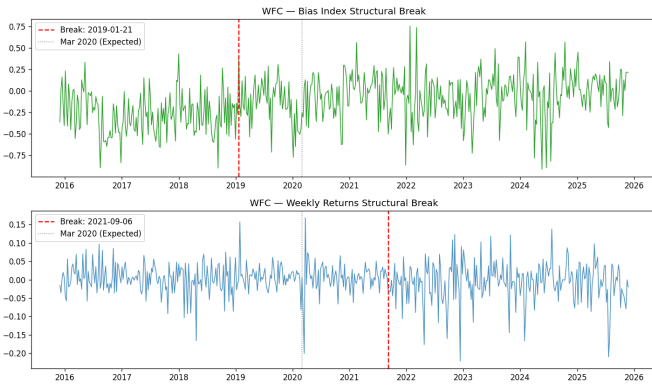


Fig. 7. Bai-Perron structural breaks for Wells Fargo (WFC), one of two firms with a cointegrating bias-return relation; its return-to-bias feedback emerges only post-break.

D. Stage 6: Structural Break Testing

Because the central hypothesis concerns a regime change around 2020, we date breaks empirically rather than imposing the calendar. A Bai-Perron procedure, implemented by dynamic programming for a single change point under an L_2 (mean-shift) cost, locates the largest structural shift in the mean of each weekly series. The estimated bias-index break dates serve as the `is_post_break` cut used to split the sample for the subsequent subsample Granger tests, and every firm yields a valid split with at least 60 observations on each side. Figures 6 and 7 show the data-driven breaks for two firms; several cluster near the COVID window while others reflect firm-specific events, which is itself evidence that a single imposed 2020 dummy would be too blunt.

E. Stage 7: VAR Specification

For each firm the VAR order is selected by estimating Eq. (6) for $p = 1, \dots, 8$ and minimizing the Bayesian Information Criterion (BIC), which penalizes parameters more heavily than AIC and so favors parsimonious, well-identified models; when BIC selects zero lags we impose $p = 1$ to retain a dynamic model. The selected orders range from 1 to

TABLE V
FIRM-LEVEL GRANGER CAUSALITY, FULL SAMPLE (HAC p -VALUES)

Firm	Lag	Bias→Ret	Ret→Bias
UBER	6	0.003	0.375
BA	5	0.097	0.119
GS	3	0.432	0.004
<i>Other 12 firms: no direction significant at 5%.</i>			

6 (modal value $p = 3$). Three residual diagnostics certify each fit. Stability requires all roots of the companion matrix to lie outside the unit circle, equivalent to $\det(\mathbf{I} - \sum_{\ell} \mathbf{A}_{\ell} z^{\ell}) \neq 0$ for $|z| \leq 1$; every selected model is stable (minimum companion root > 1.19). The Portmanteau test checks for residual autocorrelation, confirming the chosen lag is adequate for most firms. The Jarque-Bera test rejects normality everywhere, the familiar heavy-tail signature of financial data, which leaves VAR coefficients consistent but motivates robust inference; Engle's ARCH test detects volatility clustering in several return residuals, reinforcing the use of HAC-robust standard errors in the causality stage.

F. Stage 8: Granger Causality

Granger causality operationalizes “does past X help predict Y beyond Y 's own past.” Stance Granger-causes returns if the lagged stance coefficients are jointly nonzero in the return equation, i.e. $H_0 : a_1^{(2,1)} = \dots = a_p^{(2,1)} = 0$ is rejected. We test this with a direct OLS regression of each variable on p lags of both variables, using a joint Wald (F) test and Newey-West HAC-robust standard errors (`maxlags = p`) to neutralize the heteroscedasticity and autocorrelation found in Stage 7. Table V reports the firm-level full-sample p -values. Causality is sparse in the full sample, as expected if the relationship is regime-dependent: stance Granger-causes returns for Uber ($p = 0.003$) and weakly for Boeing ($p = 0.097$), while the reverse direction (returns to bias) is significant for Goldman Sachs ($p = 0.004$).

The more informative test splits each firm at its empirical bias break and re-estimates causality on each side. Several firms that are quiet in the full sample show post-break causality that is absent pre-break: Uber is the clearest, with stance Granger-causing returns at $p < 0.001$ post-break versus $p = 0.48$ pre-break, and Intel marginally so ($p = 0.074$ post). Feedback (returns→bias) emerges post-break for AT&T ($p = 0.015$), Boeing ($p = 0.010$), and Wells Fargo ($p = 0.030$), while Goldman Sachs and Morgan Stanley show it only pre-break. This pre/post asymmetry is the firm-level analogue of the structural shift the regressions probe: the media-market link is not constant but switches on after the break for a subset of firms.

G. Panel VAR

Firm-by-firm tests are underpowered because each uses only one short series. We therefore pool the cross-section into a panel VAR, which is the design the study most relies on for its dynamic conclusion. Unobserved firm heterogeneity

TABLE VI
PANEL VAR GRANGER CAUSALITY (15 FIRMS,
HELMERT-TRANSFORMED, HAC)

Sample	N	Bias→Ret	Ret→Bias
Full	7493	0.565	0.899
Pre-break	4415	0.301	0.861
Post-break	3018	0.905	0.937

is removed with the Arellano–Bover/Helmert transformation (forward mean-differencing), which subtracts from each observation the mean of all its *future* values within the firm,

$$\tilde{y}_{it} = c_{it} \left(y_{it} - \frac{1}{T_{it}} \sum_{s>t} y_{is} \right), \quad (8)$$

with a scaling factor c_{it} that equalizes the variance of the transformed errors. Unlike ordinary within-demeaning, this transformation preserves the orthogonality between the transformed variables and the lagged regressors, avoiding the look-ahead bias (Nickell bias) that contaminates dynamic fixed-effects panels. The endogenous variables are again Δ bias and the weekly log return; we additionally partial out the exogenous controls Post_t , VIX, the S&P 500 weekly log return, and the article count N_{articles} , so that any detected causality cannot be attributed to market-wide volatility or coverage volume. With $p = 3$ lags and HAC-robust standard errors, the pooled model is estimated by OLS on the transformed, constant-free data, and Granger directions are tested with joint Wald statistics over the full, pre-break, and post-break samples. Table VI reports the results.

Pooling buys power but, even so, neither Granger direction is significant at conventional levels in any subsample once VIX, the index return, and article volume are controlled. The honest reading is that the firm-level post-break signals (notably Uber, with scattered feedback effects elsewhere) do not aggregate into a uniform panel-wide media-to-returns channel: the dynamic link, where it exists, is firm-specific and heterogeneous rather than a market-wide regularity. This strengthens rather than weakens the contribution, because the panel VAR is exactly the test that would manufacture a spurious aggregate effect if the controls were omitted, and it declines to do so.

H. Limitations of the VAR Pipeline

Several caveats bound the dynamic findings. A weekly frequency, chosen for coverage, may be too coarse to capture the intraday or daily horizon over which news is plausibly priced, so a true fast media-to-price channel could be aliased away by aggregation. The single-break Bai–Perron design dates exactly one change point per series; firms with multiple regime shifts are only partially described, and the break date is estimated with sampling error that the subsample split treats as known. The bivariate VAR omits other drivers of both stance and returns (earnings surprises, analyst actions, sector shocks), so the controls in the panel model mitigate but do not eliminate omitted-variable concerns. Heavy-tailed, volatility-clustered residuals mean small-sample inference leans on the

HAC correction and asymptotics rather than exact distributions. Finally, Granger causality is predictive, not structural: a significant test says past stance forecasts returns, not that stance mechanically moves prices, and the firm-specific nature of the surviving signals counsels caution against a universal causal claim.

VII. IDENTIFICATION, VALIDITY, AND ROBUSTNESS

Three threats bear on the level-shift findings. First, macro shocks move both coverage and prices; the day and month fixed effects absorb common shocks, and the S&P 500 return and VIX controls strip the systematic component out of returns. Second, article volume differs widely across firms (Fig. 1); we hold it fixed with the volume covariate, and substituting $\ln(1 + \text{vol})$ for raw volume leaves the RQ1 estimates essentially unchanged. Third, the pre/post design assumes within-firm parallel trends; the flat pre-shock coefficients in the RQ1 monthly event study and the tightly clustered pre-period return means (range 0.0019) are consistent with this premise, though they do not prove it. We further confirm the RQ1 conclusion is robust to the stance confidence threshold ($\tau = 0, 0.6, 0.9$) and certify the estimator with the standard battery (Breusch–Pagan, Jarque–Bera, VIF, Durbin–Watson), which motivates the HC1 and day-clustered inference reported throughout. Two limitations remain: the stance signal inherits measurement noise from the upstream NewsMTSC and relevance stages, which attenuates coefficients toward zero, and a binary post indicator cannot separate the 2020 shock from other contemporaneous regime changes.

VIII. CONCLUSION

We assembled a target-dependent media-stance signal for 26 S&P 500 firms from 690,586 relevance-filtered articles and tested whether the 2020 shock moved stance, returns, and the link between them. The level-shift results are null: once common daily and monthly shocks are absorbed by two-way fixed effects, neither media stance (RQ1, $\beta = +0.014$, $p = 0.30$) nor firm-idiosyncratic returns (RQ2, $\beta = -0.002$, $p = 0.90$) show a significant aggregate shift around January 2020. The dynamic analysis (RQ3) reinforces this reading: a weekly panel VAR finds no market-wide Granger link between stance and returns in any subsample once volatility, the index return, and coverage volume are controlled, with the surviving predictability confined to a few firms (notably Uber, with scattered feedback elsewhere) rather than the panel as a whole. Taken together, the evidence points to a media–market relationship that is firm-specific and heterogeneous rather than a uniform post-2020 regularity—a conclusion that the controlled panel design supports precisely because it declines to manufacture a spurious aggregate effect.

APPENDIX

Table VII reports the decision-threshold sweep for the chosen single model on the held-out test set, which underlies Figure 3. Precision and recall trade off monotonically with

TABLE VII

HELD-OUT THRESHOLD SWEEP FOR THE CHOSEN SINGLE MODEL
($n = 190$). THE ADOPTED OPERATING POINT IS SHOWN IN BOLD.

Threshold	P	R	F1
0.50	0.878	0.857	0.867
0.70	0.875	0.833	0.854
0.75	0.897	0.833	0.864
0.80	0.919	0.810	0.861
0.85	0.914	0.762	0.831
0.90	0.912	0.738	0.816
0.92	0.968	0.714	0.822
0.95	1.000	0.714	0.833

TABLE VIII

PER-FIRM COVERAGE IN THE MODELING PANEL (RELEVANCE-FILTERED)

Ticker	Avg. Art./Day	Zero Days (%)	Weekly Cov. (%)
ABNB	0.472	78.7	68.6
AMZN	1.966	42.6	97.4
T	3.340	29.1	98.8
BA	1.334	53.7	95.3
BAC	0.357	81.7	71.3
GM	2.012	50.0	96.2
GS	0.866	60.5	94.6
INTC	1.205	58.6	92.9
MCD	0.705	73.1	79.8
MSFT	2.817	34.1	99.0
MS	0.813	60.0	95.5
SBUX	0.705	75.8	73.0
UBER	2.031	56.2	89.2
V	0.996	56.5	94.1
WMT	0.977	64.9	87.8
WFC	0.875	75.8	70.9

the threshold, and a value near 0.80 is the operating point that satisfies both targets.

Table VIII reports per-firm activity for the sixteen firms carried into the weekly modeling panel, on the relevance-filtered corpus. After relevance filtering the median firm posts well under one relevant article per trading day, which is what drives the move to a weekly frequency; at the weekly level coverage is adequate (median 92.9% of weeks) though several firms (Airbnb, Bank of America, Wells Fargo) sit near 70%. Sparse names excluded upstream (e.g. MMM, AAPL, ETN, NDAQ, NKE, ORCL, PGR, TSLA, FDX) are omitted.

Weekly gaps are classified by length: Type A (1–2 weeks) are forward-filled and Type B (3–8 weeks) linearly interpolated, while Type C (> 8 weeks) are left missing. On the relevance-filtered series 1,131 weekly observations are imputed (822 forward-fill, 309 interpolation); every firm still clears the $\geq 60\%$ coverage threshold with no gap exceeding eight weeks (Table IX). The coverage map for the sparsest retained firm is shown in Fig. 4.

Weekly log returns are computed from Monday-anchored ISO weeks and aligned to the bias index (Table X). Extreme weekly moves ($> \pm 15\%$) cluster in the March 2020 crash window. Returns are winsorized at ± 5 standard deviations from

TABLE IX
WEEKLY GAP CLASSIFICATION AND IMPUTATION COUNTS

Ticker	Cov. (%)	Gaps A	Gaps B	Longest	Imputed
ABNB	68.6	84	18	7	180
AMZN	97.4	11	1	4	15
T	98.6	1	1	7	8
BA	95.3	27	0	1	27
BAC	71.1	104	10	5	166
GM	96.0	20	0	2	23
GS	94.6	29	0	2	31
INTC	92.9	34	2	3	41
MCD	79.8	82	4	6	116
MSFT	98.8	7	0	1	7
MS	95.3	27	0	1	27
SBUX	73.0	89	12	7	155
UBER	89.0	41	2	8	63
V	93.9	25	1	7	35
WMT	87.8	52	2	8	70
WFC	70.9	69	20	8	167

each firm's mean, with AT&T (T) excluded from winsorization so that its outlier behavior can be compared against the others.

Integration orders combine the ADF and KPSS verdicts on the levels and first differences of each series (Table XI). Returns are $I(0)$ almost everywhere; on the relevance-filtered series the bias index is now $I(0)$ in levels for about half the firms and $I(1)$ or break-driven for the rest. A Johansen trace test detects cointegration for two firms, Boeing (trace = 367.23) and Wells Fargo (trace = 317.53), both far above the 95% critical value of 15.49; these are the VECM candidates noted as robustness exceptions to the uniformly differenced specification.

Lag orders are chosen by BIC over $p = 1, \dots, 8$; Table XII reports the AIC and BIC choices, the selected order, and residual diagnostics. Every selected model is stable (minimum companion root > 1). Jarque–Bera rejects normality everywhere (heavy tails), and Engle's ARCH test flags volatility clustering in most return residuals, together motivating the HAC-robust causality tests. WMT is excluded from this specification pass.

Bai–Perron single-break dates for each weekly series, with the resulting pre/post observation counts, are in Table XIII. The bias-index break defines the subsample split for the causality tests. General Motors breaks at 2020-01-13, closest to the COVID shock; most other bias breaks are firm-specific.

Table XIV reports HAC Granger p -values for every firm, in both directions, over the full sample and the pre/post subsamples defined by the bias break. Stars denote $*p < 0.10$, $**p < 0.05$, $***p < 0.01$.

Table XV summarizes the classical-linear-model diagnostics for the RQ1 two-way fixed-effects estimator. Orthogonality and conditioning are satisfied; heteroscedasticity and non-normality are present and handled by the HC1/clustered errors and large-sample asymptotics, respectively; collinearity is negligible.

TABLE X
WEEKLY LOG-RETURN QUALITY AND WINSORIZATION (16 FIRMS)

Ticker	Weeks	Mean	Std	Min	Max	Extreme ($> \pm 15\%$)	Winsorized
ABNB	522	0.00265	0.03562	-0.1443	0.1399	0	0
AMZN	522	0.00235	0.03230	-0.1829	0.1138	1	1
T	522	0.00147	0.06861	-0.4093	0.3331	18	skipped
BA	522	-0.00046	0.03924	-0.2849	0.1939	4	1
BAC	260	-0.00090	0.06274	-0.1815	0.1896	5	0
GM	522	0.00503	0.05559	-0.2923	0.1868	6	1
GS	522	0.00298	0.03935	-0.2026	0.1322	2	1
INTC	522	0.00224	0.03584	-0.1571	0.1397	1	0
MCD	522	0.00251	0.04381	-0.2056	0.2160	5	0
MSFT	522	0.00209	0.04499	-0.2197	0.1940	6	0
MS	522	0.00335	0.04431	-0.2955	0.2533	6	2
SBUX	522	0.00264	0.04417	-0.1883	0.1934	3	0
UBER	522	0.00250	0.04123	-0.2416	0.2012	4	1
V	522	0.00154	0.05526	-0.2859	0.1954	6	1
WMT	522	0.00154	0.05526	-0.2859	0.1954	6	1
WFC	522	0.00012	0.04667	-0.2215	0.1680	10	0

TABLE XI
INTEGRATION ORDER OF BIAS AND RETURN SERIES

Ticker	(Bias, Return)	Note
ABNB	$(I(0), I(0))$	both stationary
AMZN	$(I(1), I(0))$	
T	$(I(0), I(0))$	both stationary
BA	$(I(1), I(1))$	
BAC	$(I(1), I(0))$	cointegrated (VECM)
GM	$(I(0), I(0))$	both stationary
GS	$(I(1), I(0))$	
INTC	$(I(0), I(0))$	both stationary
MCD	$(I(0), I(0))$	both stationary
MSFT	$(I(1), I(0))$	both stationary
MS	$(I(0), I(0))$	
SBUX	$(I(1), I(0))$	both stationary
UBER	$(I(1), I(0))$	
V	$(I(0), I(0))$	both stationary
WMT	$(I(0), I(0))$	both stationary
WFC	$(I(1), I(1))$	cointegrated (VECM)

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TABLE XII
VAR LAG SELECTION AND RESIDUAL DIAGNOSTICS (15 FIRMS)

Ticker	AIC	BIC	Chosen	N	Min Root	Portm. p	JB p	ARCH(B) p	ARCH(R) p
ABNB	5	3	3	521	1.62	0.0356	0.0000	0.0097	0.0679
AMZN	7	2	2	521	1.93	0.0002	0.0000	0.0202	0.0000
T	7	6	6	521	1.24	0.2131	0.0000	0.2725	0.7046
BA	8	5	5	521	1.29	0.2558	0.0000	0.0120	0.0000
BAC	5	1	1	259	3.09	0.2267	0.0000	0.8870	0.2809
GM	8	4	4	521	1.39	0.0416	0.0000	0.0662	0.0004
GS	7	3	3	521	1.48	0.0038	0.0000	0.7429	0.0003
INTC	8	3	3	521	1.58	0.0139	0.0000	0.0363	0.0000
MCD	8	3	3	521	1.56	0.0016	0.0000	0.5908	0.0000
MSFT	8	3	3	521	1.53	0.0114	0.0000	0.0134	0.2055
MS	8	4	4	521	1.42	0.0024	0.0000	0.0645	0.0000
SBUX	6	3	3	521	1.55	0.0475	0.0000	0.7787	0.0000
UBER	8	6	6	521	1.19	0.0042	0.0000	0.0001	0.0000
V	7	3	3	521	1.56	0.0033	0.0000	0.0032	0.0000
WFC	8	1	1	521	2.57	0.0003	0.0000	0.0410	0.0000

TABLE XIII
EMPIRICAL STRUCTURAL BREAK DATES

Ticker	Bias Break	Return Break	Pre	Post
ABNB	2023-02-27	2021-09-06	378	144
AMZN	2018-10-22	2022-01-03	151	371
T	2022-11-28	2021-09-27	365	157
BA	2019-03-04	2022-01-03	170	352
BAC	2022-05-02	2022-12-26	73	187
GM	2020-01-13	2017-10-30	215	307
GS	2024-01-15	2021-11-22	424	98
INTC	2024-07-22	2023-03-20	451	71
MCD	2020-06-29	2023-10-30	239	283
MSFT	2022-05-30	2021-09-27	339	183
MS	2017-07-03	2023-10-30	83	439
SBUX	2022-02-07	2022-10-03	323	199
UBER	2023-01-16	2024-04-01	372	150
V	2024-06-10	2020-03-23	445	77
WMT	2024-04-22	2020-03-23	438	84
WFC	2024-10-07	2021-09-06	462	60

TABLE XIV
FIRM-LEVEL GRANGER CAUSALITY, FULL AND SUBSAMPLE (HAC p -VALUES)

Ticker	Lag	Full		Pre-break		Post-break	
		B→R	R→B	B→R	R→B	B→R	R→B
ABNB	3	0.362	0.892	0.165	0.884	0.416	0.936
AMZN	2	0.944	0.277	0.228	0.795	0.772	0.137
T	6	0.570	0.367	0.525	0.670	0.484	0.015**
BA	5	0.097*	0.119	0.091*	0.247	0.258	0.010**
BAC	1	0.945	0.550	0.578	0.407	0.778	0.849
GM	4	0.629	0.543	0.805	0.515	0.498	0.519
GS	3	0.432	0.004***	0.397	0.026**	0.919	0.210
INTC	3	0.611	0.363	0.668	0.535	0.074*	0.157
MCD	3	0.276	0.518	0.178	0.456	0.920	0.461
MSFT	3	0.537	0.847	0.157	0.060*	0.981	0.721
MS	4	0.389	0.959	0.019**	0.514	0.462	0.938
SBUX	3	0.419	0.207	0.187	0.286	0.682	0.542
UBER	6	0.003***	0.375	0.478	0.270	0.000***	0.131
V	3	0.281	0.860	0.253	0.871	0.854	0.268
WFC	1	0.303	0.148	0.593	0.497	0.138	0.030**

TABLE XV
RQ1 SUMMARY DIAGNOSTICS

Test	Statistic	Implication
Residual mean	-9.0×10^{-19}	orthogonality OK
Condition number	157.46	numerically stable
Breusch–Pagan (LM)	190.25	heteroscedastic (HC1 used)
Jarque–Bera	4541.18	fat tails (n large)
VIF (both regressors)	≈ 1.00	no collinearity
Max pairwise corr.	0.0189	negligible
Durbin–Watson	1.673	mild autocorrelation